

Field and landscape risk factors impacting Flavescence dorée infection : Insights from spatial Bayesian modelling in the Bordeaux vineyards

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Abstract

Flavescence dorée (FD) is a quarantine disease threatening European vineyards. Its management is based on mandatory insecticide treatments and the uprooting of infected plants identified during annual surveys. Field surveys are currently not optimised, as the drivers affecting FD spread in vineyard landscapes remain poorly understood. We collated a georeferenced dataset of FD detection, collected from 34581 vineyard plots over 5 years in South-West France. Spatial models fitted with INLA (Integrated Nested Laplace Approximation) were used to identify local and landscape factors affecting FD detection and infection. Our analysis highlights the importance of sampling period on FD detection and of local practices and landscape context on FD infection. At field scale, altitude and cultivar choice were the main factors affecting FD infection. In particular, the odds ratio of FD infection in fields planted with the susceptible Cabernet Sauvignon, Cabernet Franc or Muscadelle varieties were about twice that in fields planted with the less susceptible Merlot. Field infection was also affected by the field's immediate surroundings (in 150 – 200 m radius circle), corresponding to landscapes of 7 – 12 ha. In particular, the probability of FD infection increased with the proportions of forest and urban land and with the proportion of susceptible cultivars, demonstrating that the cultivar composition impacts FD epidemiology at landscape scale. The satisfactory predictive performance of the model for identifying districts with prevalence of FD detection exceeding 10% of the fields suggests that it could be used to target areas in which future surveys would be most valuable.

keywords— Landscape epidemiology, vineyard disease, Distribution modelling, INLA, Varietal landscape, Landscape mosaic

Introduction

An understanding of the contribution of environmental variables to the presence and spread of any pathogen is crucial for designing efficient surveillance and management disease strategies. Species distribution models are used for this purpose in epidemiology (Purse and Golding 2015, Bebbier 2015). These correlative models can be used to disentangle the relative effects of multiple environmental variables (e.g., biotic and abiotic factors) on pathogen epidemiology. They also make it possible to make predictions, for all sites in a region of interest for which the factors studied have been mapped. It therefore provides a basis for the implementation of targeted surveillance and improvements in disease control, by maximising the detection of new cases (Parnell et al. 2014). Widely used for the mapping and management of infectious diseases in humans (Kraemer et al. 2016), distribution modelling has also been used to obtain information about the spatial distribution of emerging plant diseases, such as sudden oak disease in the USA (Meentemeyer et al. 2008, Vaclavik et al. 2010), citrus black spot in South

Africa (Martinez-Minaya et al. 2018) and *Xylella fastidiosa* in Europe (Godefroid et al. 2019, Cendoya et al. 2020).

On the methodological side, the first step in any species distribution modelling approach is to gather together diverse sources of information about potential risk factors (e.g., host characteristics, cropping practices, climatic data, and land use) and georeferenced records of the presence of the pathogen (Purse and Golding 2015, Kraemer et al. 2016). The next step involves the use of statistical tools to investigate correlations between these variables (Elith and Leathwick. 2009, Dormann et al. 2021). This step is not straightforward as the analysis of spatial data is often complicated by a phenomenon known as spatial autocorrelation (Dormann et al. 2007). Spatial autocorrelation occurs when the values of variables sampled at nearby locations are not independent from each other. In such a case, one of the key assumptions of standard statistical analyses, which is that residuals are independent and identically distributed, is violated. The violation of this assumption may bias parameter estimates and can increase type I error rates (falsely rejecting the null hypothesis of no effect) (Dormann et al. 2007). A variety of frequentist and Bayesian statistical approaches have been developed over the last decades for modelling spatial data while accounting for spatial autocorrelation (Dormann et al. 2007, Benguin et al. 2012). Among them, Integrated nested Laplace approximation (INLA) (Rue et al. 2009, Lindgren et al. 2011) has emerged in the last decade as a highly appealing alternative combining both outstanding computational speed and availability in a user-friendly R interface in the R-INLA library (Zuur et al. 2017, Benguin et al. 2012).

Phytoplasmas are plant-pathogenic bacteria, pleiomorphic with no cell walls, that belong to the class of Mollicutes (Namba 2019). They are obligate parasites invading the phloem sieve tube elements of the host plants and colonizing the body of insect vectors. Phytoplasmas are transmitted by phloem feeding hemipteran insects (leafhoppers, planthoppers and psyllids) (Weintraub and Beanland 2006) and by vegetative propagation of infected plant material. They are associated with diseases that cause severe economic impacts on many crops worldwide (Namba 2019). Flavescence doree (FD), one of the most damaging diseases in European vineyards, is caused by the Flavescence dorée phytoplasma (taxonomic subgroups 16SrV-C and 16SrV-D). FD phytoplasma is transmitted from grapevine to grapevine by the leafhopper vector *Scaphoideus titanus* (Chuche and Thiéry 2014). Typical symptoms are leaf yellowing or reddening, with downward rolling, incomplete lignification of canes, abortion of flowers, and grape wilting. FD disease emerged in South-West France in the 1950s, following the accidental introduction of *S. titanus* from North America (Caudwell 1957, Papura et al. 2012). But FD phytoplasma was demonstrated to be European, originating from wild plant reservoirs: the alder tree *Alnus glutinosa* (Fagales, Betulaceae) and the climbing shrub *Clematis vitalba* (Ranunculales, Ranunculaceae) from which the phytoplasma was originally transmitted to cultivated grapevines (Malembic-Maher et al. 2020).

At the field scale, the epidemiology of FD is firstly driven by the dynamics of its ampelophagous vector *S. titanus*. This species is univoltine. The eggs hatch in April on grapevine, and there are then five nymphal instars before the first adults appear, usually in June and July. The adults live for about one month. The fertilised females lay eggs in the late summer, from August to September. Phytoplasmas are acquired passively, from the first larval stage onwards, through feeding of the phloem sap of infected grapevine plants where the phytoplasma multiply. Once infected, the insects carry and transmit the phytoplasma for the rest of their lives (Weintraub and Beanland 2006). The flight activity of the vector depends on vine density and canopy architecture (Lessio and Alma. 2004, Lessio et al. 2015). The epidemiology of FD is also driven by the differential propensity of vine cultivars to act as a source of inoculum (Bressan et al. 2005, Galetto et al. 2014). No cultivars are resistant to FD. They rather display a continuum from low (e.g. Merlot) to high (e.g. Cabernet-Sauvignon) susceptibility to the presence and multiplication of phytoplasma (Eveillard et al. 2016, Ripamonti et al. 2021, Oliveira et al. 2019). Once the disease has become chronically installed in a region, *Vitis* spp. growing in abandoned vineyards or close to cultivated vineyards can constitute an important source of primary infection (Tramontini et al. 2020). Indeed, they provide a reservoir for both the phytoplasma and its vector (Lessio et al. 2014, Ripamonti et al. 2020). Although dispersal abilities of the vector is rather small (< 30 m), longer range dispersals have been observed, sug-

gesting that landscape structure may also affect the epidemiology of FD (Lession et al. 2014).

In the last decades, FD has spread throughout European vineyards (Jeger et al. 2016). Due to the severe economic consequences of the disease, FD phytoplasma is classified as a quarantine organism in Europe since 1993. There is currently no means of curing plants of FD phytoplasma. The disease is therefore controlled principally by four mandatory measures: (i) the planting of disease-free material, (ii) the application of insecticides to kill the vector, (iii) the establishment of annual vineyard surveys for monitoring plant infection and (iv) the uprooting of infected plants. Two factors complicate the detection of FD in the field. First, FD cocirculates in European vineyards with Bois noir (BN), another phytoplasma disease that has similar symptoms but does less economic damage (Quaglino et al. 2013, Tramontini et al. 2020). Molecular tests involving the detection of pathogen DNA in a real-time multiplex PCR assay are therefore required to confirm the presence of FD (Pelletier et al. 2009). Second, typical FD symptoms appear only during the summer of the year following inoculation (Schvester et al. 1969, Morone et al. 2007, Tramontini et al. 2020). Plants newly infected during the spring and early summer of year n therefore constitute a source of inoculum before the symptoms of the disease become visible, late in the summer of year $n + 1$, when the monitoring campaigns are performed. Once the diagnosis of FD is confirmed, the infected plants must be removed and insecticide treatments are mandatory within defined perimeters, with no financial compensation for the growers. More than 70% of French vineyards are treated with insecticides against *S. titanus*, with consequences for both the environment and human health (Desneux et al. 2007, Tang et al. 2021). Surveillance has been intensified over the last decade under the supervision of legal authorities, leading to a reduction of the number of annual treatments.

In this study, we applied a distribution modelling approach to improve our understanding of FD epidemiology and, more specifically, gain a quantitative view of the field and landscape factors impacting the probabilities of FD detection and infection at field scale. Based on the literature, we first hypothesized that the probability of FD detection improves as fall approaches. We also hypothesized that key local factors would impact the probability of FD infection with a higher probability of infection associated to older and denser plantation as well as cultivation of more susceptible cultivars. We also hypothesized that the probability of infection would increase with the proportion of susceptible cultivars, the proportion of semi-natural habitats or the proportion of urban areas in the surrounding landscape as these habitats may act as FD reservoirs. In addition, we hypothesized that higher level of fragmentation of these potential reservoirs at the landscape scale would limit vector spillover and therefore be associated with lower probability of infection (Tschardt and Brandl 2004). To test these hypotheses, we fitted distribution models using INLA to a spatial dataset collected from 34581 vineyard fields over 5 years of mandatory vineyard surveys. Finally, we investigated whether the model could identify sites with higher probability of infection for targeted surveillance to improve current management strategies.

Materials and methods

Flavescence dorée monitoring and detection

In the vineyards of Bordeaux in South-West France, FD surveys are performed by professional organisations known as GDONs (Groupements de Défense contre les Organismes Nuisibles), under the supervision of the French Ministry of Agriculture. The "GDON des Bordeaux" has been in charge of FD monitoring since 2011 in an area of 364718 ha of which 83912 ha (23%) were cultivated with grapevine in 2016 according to the "Casier Viticole Informatisé" (CVI). CVI is a GIS database created by the French directorate general of customs which provides a comprehensive history of each legal piece of land cultivated with grapevine. It gathers information on the spatial boundary, size, year and density of plantation and cultivars used in each legal field. The region monitored by GDON is subdivided into 347 "districts", with a mean area of 1048 ha and 10% and 90% quantiles of 331 ha and 1849 ha, respectively (Fig. 1A). Although these districts serve as a basis for the practical organization of FD monitoring performed by the GDON in the vineyards, the basic observational unit is the field. In the

155 following, we considered the annual survey realized from 2012 to 2016. Each year, the monitoring
 156 strategy of GDON was nearly identical. It was conducted by a team of trained inspectors between
 157 August and October, the period of the year most favourable for the detection of FD symptoms
 158 (Tramontini et al. 2020). The fields selected each year n by the GDON can be separated into 2
 159 categories. The first category consists of the fields already surveyed and detected infected the
 160 previous year $n - 1$. Once an infected field is detected, GDON inspectors check the following year
 161 that mandatory control measures were applied. These re-inspection data were not considered
 162 in our study. The second category consists of the fields not surveyed in the previous years.
 163 Approximately 10% of the vineyard area was newly inspected each year. These fields were
 164 considered in our study. The survey involved the inspection of most of the fields in large parts
 165 of newly selected districts each year, most often regardless of prior information on the presence
 166 of FD (note that “spontaneous” reporting of FD by the winegrowers is unusual). Overall a mean
 167 of 6916 fields are newly inspected each year, a number ranging from 5916 in 2013 to 8712 in
 168 2012. The fields had a mean area of 0.77 ha, with 10% and 90% quantiles of 0.16 ha and 1.6
 169 ha, respectively.

170 When the inspection teams survey a given area, they inspect a high proportion of fields. We
 171 quantify this sampling effort by the proportion of the vineyard sampled by GDON inspectors.
 172 For each of the 34581 fields sampled, the sampling effort was assessed by the ratio between (i)
 173 the total vineyard area sampled by GDON inspectors in a radius of 1 km around each field (as
 174 assessed from the area of the polygons drawn by inspectors during their survey, see below) and
 175 (ii) the total size of the vineyard in this same radius (as given by the area of legal piece of land
 176 cultivated with grapevine provided by the CVI). The mean sampling effort was 0.58, with 10%
 177 and 90% quantiles of 0.38 and 0.74, respectively. As previously stated, the basic observational
 178 unit was the field (Fig. 1B). For each field surveyed, inspectors walk through the whole field for
 179 visual detection of the symptoms and draw its polygon on an GIS software. If no symptoms are
 180 detected, the inspectors do not sample any plants and the field is declared as “uninfected”. If
 181 symptoms are detected, symptomatic plants are marked for uprooting and symptomatic leaves
 182 from one to five plants per field are collected and pooled into a single sample. The detection of the
 183 FD and BN phytoplasmas in samples is performed by accredited laboratories, with a real-time
 184 PCR triplex molecular test derived from that described by (Pelletier et al. 2009). Each field is
 185 then classified as either infected with FD (if the molecular test is positive for FD phytoplasma,
 186 meaning that at least one of the plant in the sample is infected with FD) or uninfected with
 187 FD. Overall, FD was detected in 7.6% of the 34581 fields surveyed and BN detected in 4.5%.

188 Explanatory variables at field scale

189 In addition to the georeferenced records of FD detection, we collected a set of potentially impor-
 190 tant local explanatory variables characterizing the production situations of each field (Table 1).
 191 First, the **altitude** was obtained from topography data provided by IGN (Institut national de
 192 l’information géographique et forestière). Second, the **AOC** (Appellation d’Origine Contrôlée),
 193 a French certification defining the geographic area of production for particular wine label and
 194 common and specific guidelines (e.g. proportions of particular cultivars, growing practices, wine-
 195 making practices, etc.) was noted, to provide information about the socioeconomic context of
 196 production. The AOCs were grouped into six levels on the basis of geographic proximity. Third,
 197 the CVI was used to determine, for each field, the **cultivars** grown (7 levels, Fig. 1 B), plant
 198 **density** and the **age** of the plantation. The protocol for extracting these variables from the CVI
 199 is described in Supplementary Material (Text S1, Table S1). Fourth, the type of viticultural
 200 **Practice** (organic or conventional) in each field was provided by the GDON. This information
 201 was available only for 2016, but we assumed that growers maintained the same practices over
 202 the entire time period considered here.

203 Explanatory variables at landscape scale

204 We created annual land covers maps of the region monitored by GDON des Bordeaux. These
 205 maps are raster maps with a 10-metre resolution. The maps were created by overlaying the CVI
 206 database and the France land cover map for 2017 (OSO map, (Inglada et al. 2017)). We used

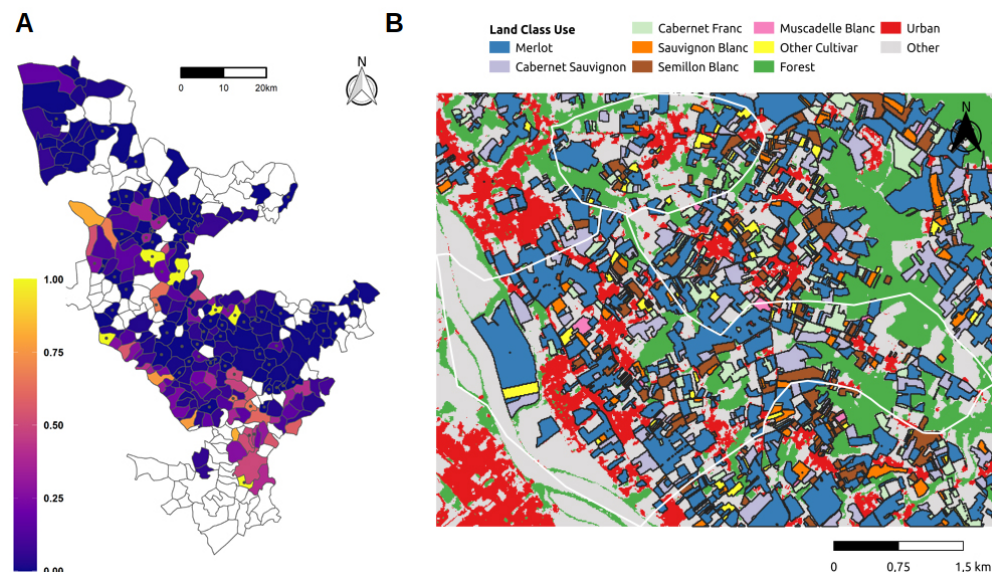


Figure 1: Field-based Flavescence dorée (FD) data at the regional and district scales. A: Map of the study region in South-West France. The map shows the borders of the 347 districts located within the GDON des Bordeaux area. The colour scale indicates the prevalence of FD detection per district in each of the 239 districts in which FD surveys were performed from 2012 to 2016. Unserved districts are shown in white and districts with less than 10 fields are shown with a dot. B: Maps at landscape scale (over 2 entire districts depicted by white border) illustrating the vineyard plots, classified according to their cultivar (seven levels), and the other categories of land use (three levels).

207 the OSO map for 2017 because (i) the landscape was expected to be constant within the time
208 frame of the study and (ii) the map resolution was more precise in 2017 (pixels of 10 m per side)
209 than previous years (pixels of 20 m per side). For each year from 2012 to 2016, the CVI database
210 was used to map the pixels representing grapevine cultivars. Seven classes depicting the cultivar
211 used in each pixel during the year considered were distinguished (as listed in Table 1) while
212 an independent eighth class indicates whether the pixel belongs to an organic or conventional
213 vineyard farm (data provided by GDON, see previous section). Each pixel of the GDON region
214 not attributed to a vineyard class was classified into 3 separate classes (forest, urban or other
215 land use) depending on its classification in the OSO map (Fig. 1B). The resulting land cover
216 map for 2016 is illustrated in Figure S1.

217 These annual land cover maps were used to characterize the landscape surrounding each
218 field. We calculated composition and configuration metrics in circles of 15 increasing radius (50,
219 100, 150, 200, 250, 300, 500, 1000, 1500, 2000, 2500, 3000, 4000, 5000 and 6000 m) centered
220 on the barycentre of each field i ($i = 1, \dots, 34581$) (Table 1). Thereafter, “scale” defined the
221 radii around field barycentres (*i.e.* the extent of the landscapes). For each field i and scales
222 considered, we calculated the percentage of land occupied by vineyard **L_vineyard** (whatever
223 the cultivar grown), forest **L_forest** and urban areas **L_urban**. These metrics are assessed as
224 the proportion of pixels attributed to each class within the circle. We also calculated the total
225 number of patches **L_np** of these 3 classes, a simple measure of landscape fragmentation. A
226 patch refers here to a continuous area of the landscape attributed to a single class use. Finally,
227 we calculated the percentage of land (i) cultivated with the cultivar Merlot **L_merlot** and
228 (ii) organically managed **L_organic** within the total area devoted to grapevine cultivation
229 in the circle considered. These metrics were computed with the R package landscapemetrics
230 (Hesselbarth et al. 2019).

Table 1: Local and landscape explanatory variables used to characterize the vineyard fields.

Type	Variables	Description	Mean (sd)/levels ¹	Source
Local variables	Area	Field area	0.77 <i>ha</i> (0.77)	GDON
	Year	Year of the first field inspection	Five levels	
	Season	Summer or Autumn inspection	Two levels ²	
	Week	Week number within year	Eleven levels ²	
	Practice	Organic or conventional practices	Two levels ²	
	AOC	Appellation d'origine contrôlée	Six levels ³	INAO
	Age	Age of the plantation at inspection	23 <i>years</i> (14)	CVI
	Cultivar	Grape cultivar	Seven levels ⁴	
	Density	Density of the plantation	4131 <i>plants/ha</i> (1008)	
	Altitude	Altitude	59.5 <i>m</i> (29)	IGN
Landscape scale	L_urban	Percentage of urban area	8.2 (10)	CESBIO
	L_forest	Percentage of forest area	5.6 (9.8)	
	L_vineyard	Percentage of vineyard area	66.7 (20.2)	
	L_merlot	Percentage of merlot area in the vineyard	62.2 (32)	CVI
	L_organic	Percentage of organic practice area in the vineyard	8 (21.3)	GDON
	L_np	Number of vineyard, forest and urban patches	12.5 (5.5)	

¹ Values for the landscape covariates are given for a scale of 150m.

² Season : Summer here is the period from July August (28719 fields) and Autumn corresponds to September-October period (5862 fields). Week : 11 levels from week 32 to week 42. Practice : Organic (2745 fields) or conventional (31836 fields) farming practices.

³ The 11 controlled designation of origin located in the GDON des Bordeaux area were grouped by spatial proximity in six levels in order to have at least 700 fields in each level. These levels are AOC1 (Côtes de Bordeaux-Saint-Macaire, 1936), AOC2 (Premières Côtes de Bordeaux, 5032 fields), AOC3 (Loupjac, Sainte-Croix-du-Mont and Graves supérieures, 725 fields), AOC4 (Fronsac and Cannon Fronsac, 743 fields), AOC5 (Cotes de Bordeaux Blaye, 5858 fields) and AOC6 (Bordeaux, 20287 fields).

⁴ Cultivars include merlot (21181 fields), cabernet-sauvignon (5266 fields), cabernet-franc (2491 fields), muscadelle (270 fields), sauvignon (1985 fields), semillon (2587 fields) and other cultivars (801 fields) not previously listed.

231 Statistical analysis

232 Bayesian inference with INLA

233 Integrated nested Laplace approximation (INLA) is a computationally efficient method for fitting
 234 complex spatial models within the Bayesian paradigm (Rue et al. 2009, Lindgren et al. 2011).
 235 It offers a faster alternative to Markov Chain Monte Carlo (MCMC) methods as, in essence,
 236 INLA replaces stochastic sampling with deterministic approximation based on a clever use of the
 237 Laplace approximation and on numerical integration (Benguin et al. 2012). INLA may be used
 238 to fit a large class of latent Gaussian models in a Bayesian framework. Moreover, the method is
 239 implemented in the R package INLA (<https://www.r-inla.org/>) (Lindgren and Rue 2015) which
 240 allows fitting models almost as easily as the base R functions for generalized linear models.

241 Latent Gaussian models include in particular several classes of models accounting for spatial
 242 autocorrelations. Indeed, combined with the stochastic partial differential equation approach
 243 (SPDE), one can accommodate with INLA all kinds of geographically referenced data, includ-
 244 ing areal and geostatistical ones, as well as spatial point process data (Lindgren et al. 2011,

245 Lindgren and Rue 2015). Here, we employ INLA to fit spatial logistic regression in a contin-
 246 uous spatial domain. References such as (Benguin et al. 2012) and (Zuur et al. 2017) provide
 247 comprehensive text on the subject for applied ecologists and phytopathologists.

248 Spatial logistic regression

249 Let y_i be the FD detection status of the field i at location s_i on its year of first inspection $year_i$,
 250 i.e. the field i is classified as either infected (value 1) or uninfected (value 0). We assumed that
 251 y_i follows a Bernoulli random variable and modelled the probability π_i of FD detection in field
 252 i using a logit link function. The generic model can be written as

$$\begin{cases} y_i \sim \text{Binomial}(1, \pi_i), & i = 1, \dots, 34581 \\ \text{logit}(\pi_i) = \beta X_i + a(year_i) + u(s_i) \\ u(s) \sim \text{GMRF}(0, \mathbf{Q}^{-1}(\phi, \sigma_u^2)) \\ a(year_i) \sim \text{Normal}(0, \sigma_{year}^2) \end{cases} \quad (1)$$

253 In this model, X_i is the vector of explanatory variables for field i , and β is the vector of
 254 associated coefficients to be estimated. All the explanatory variables X_i listed in Table 1 were
 255 treated as fixed effects except the year of first field inspection treated as independent random
 256 effects $a(year_i)$. The term $u(s_i)$ is the spatial random effect for field i . It endows the model with
 257 a spatial dependence between neighbouring fields not explained by the explanatory variables. In
 258 R-INLA, a computationally convenient Gaussian Markov random field (GMRF) representation
 259 is used to accurately approximate a Gaussian random field with spatial variance and autocor-
 260 relation characterised by the Matérn covariance function (Lindgren et al. 2011). This approxi-
 261 mation relies on basis functions anchored at a set of discrete points corresponding the nodes of
 262 a mesh dividing the study area into a large number of non-overlapping triangles (Fig. S2). Its
 263 use allows inference about the latent spatial field over the entire continuous domain of interest,
 264 where the spatial interpolation between the Gaussian variables located at the nodes is linear.

265 The GMRF can be regarded as a multivariate Gaussian distribution (with dimension equal
 266 to the number of nodes) with zero mean and sparse precision matrix $\mathbf{Q}$ that depends on two
 267 positive hyperparameters σ_u^2, ϕ describing the variance and the correlation range (the distance
 268 beyond which pairs of observations are approximately not spatially correlated any more), re-
 269 spectively (Lindgren et al. 2011, Krainski et al. 2018). The spatial dependence is encoded in
 270 the matrix $\mathbf{Q}$ of size $N \times N$ by using the Matérn covariance function based on the first-order
 271 Bessel function K_1 . Accordingly, the covariance between $u(s_i)$ and $u(s_j)$ depends on the
 272 two unknown hyperparameters and on the Euclidean distance $d(i, j)$ between the two fields
 273 through $\text{cov}_{\text{Matern}}(u(s_i), u(s_j)) = \sigma_u^2 \sqrt{8}(d(i, j)/\phi) K_1(\sqrt{8}(d(i, j)/\phi))$. We remark that different
 274 parametrizations exist for the Matérn covariance, and the preceding one offers the advantage
 275 of relatively simple intuitive interpretation. The corresponding correlation function is obtained
 276 by replacing σ_u^2 by 1 in $\text{cov}_{\text{Matern}}$. In our model (1), we refer to the sparse precision matrix
 277 resulting from this covariance model as $Q(\phi, \sigma_u^2)$.

278 Bayesian inference requires to specify prior distributions for the model parameters and hy-
 279 perparameters (i.e. parameters of prior distributions). These choices are less crucial with
 280 large datasets as the effect of prior choice on the posterior estimates is expected to wane as
 281 the sample size increases. For the parameters β_j and the hyperparameter σ_{year}^2 , we used
 282 the default internal vague priors recommended in R-INLA, $\text{Normal}(0, \text{precision} = 10^{-3})$ and
 283 $\text{Log-Gamma}(1, 0.00005)$, respectively. We used penalized complexity priors (PC-priors) for the
 284 range and the variance of the spatial random effect (Simpson et al. 2017, Fuglstrad et al. 2019).
 285 PC-priors can be defined via intuitive probability statements. As few fields are less than
 286 0.25 km apart, we assume that the probability that ϕ is < 0.25 km is 0.01, leading to a
 287 $PC - \text{prior}(0.25, 0.01)$ for ϕ . Moreover, we assume that the probability that σ_u is > 1 is
 288 0.01, leading to a $PC - \text{prior}(1, 0.01)$ for σ_u .

289 Model comparison

290 The baseline spatial model M_0 corresponds to equation 1 with 13 explanatory variables, includ-
 291 ing 7 local variables *Area*, *Age*, *Practice*, *Density*, *Altitude*, *Cultivar* and *AOC* characterizing
 292 the field and 6 landscape variables $L_Organic$, $L_Vineyard$, L_np , L_Forest , L_Merlot and
 293 L_Urban characterizing the landscape within a zone of radius r . No interaction between ex-
 294 planatory variables were considered and all continuous explanatory variables were standardized.
 295 In all, 16 models M_0 corresponding to landscape scales r ranging from 0 to 6000 m were consid-
 296 ered. The case $r = 0$ corresponds to a case in which only local variables are considered. These
 297 16 spatial models were compared to their 16 non-spatial counterparts for which the spatial
 298 random effect $u(s_i)$ was removed. We estimated the Watanabe Information Criterion (WAIC)
 299 (Watanabe 2010, Gelman et al. 2014), which penalises model complexity, and the Deviance In-
 300 formation Criterion (DIC) (Spiegelhalter et al. 2014) to compare all the models fitted.

301 In addition to the baseline model M_0 , we considered a model M_1 to refine the study of the
 302 period of field inspection. In model M_1 , the variable *Season* is eliminated from the set of fixed
 303 effects X_i and replaced by a random walk of order 1 indexed by the weeks of inspection. The
 304 weeks range from week 32 (second week of august) to week 42 (third week of october). The
 305 notation k_i denotes thereafter the week k of inspection of field i ($k = 32, \dots, 42$). In model
 306 M_1 , the logit equation in 1 is replaced by $\text{logit}(\pi_i) = \beta X_i + a(\text{year}_i) + u(s_i) + w(k_i)$ with
 307 $w_k = w_{k-1} + \nu_k$ ($k = 33, \dots, 42$) and $\nu_k \sim \text{Normal}(0, \sigma_k^2)$. In this model, σ_k^2 is an additional
 308 hyperparameter for which we used the default internal INLA prior $PC - \text{prior}(0.5, 0.01)$.

309 Preliminary analysis

310 A set of preliminary analysis was realized with the baseline model M_0 (not shown). First,
 311 a test for multicollinearity was performed, by fitting the model M_0 without spatial random
 312 effect for the 16 landscape scales considered. The variance inflation factors for each variable
 313 were < 3 (except for *Practice* and $L_organic$ at the landscape scale of 50 m radius with values
 314 < 5), indicating an absence of collinearity between the explanatory variables (Zuur et al. 2009).
 315 Second, we compared the model M_0 with a binomial distribution to an alternative model re-
 316 lying on a zero-inflated binomial (ZIB) distribution, to account for a possible excess of zeros
 317 (Martin et al. 2005), as 92.5% of the fields were FD-negative. We found that the binomial
 318 models outperformed the ZIB models, suggesting the effectiveness of the surveillance. Finally,
 319 we investigated the effect of six meshes on parameter estimations. The meshes differ in the
 320 largest triangle edge length (parameter max.edge) and the minimum allowed distance between
 321 points (parameter cutoff) (Fig. S2). Overall, the parameters estimates were consistent from the
 322 coarsest to the finest meshes (Fig. ??).

323 Predictive performance of the models

324 The predictive performance of the baseline model M_0 was estimated at two scales (field, district)
 325 by cross-validation. The model was trained on 80% of the data (training set) and tested on the
 326 remaining 20% (testing set). We repeated this partitioning for cross-validation 50 times. We
 327 first assessed the ability of the model to predict the FD detection status of individual fields with
 328 precision-recall curve (PR curve). PR curve were preferred over ROC curve as they are more
 329 informative when evaluating binary classifiers on imbalanced datasets which is the case in our
 330 study with 7.5% of FD detection (Saito and Rehmsmeier 2015). For a given threshold (between
 331 0 and 1) above which a field is classified as infected by the model, the recall, also known as
 332 sensitivity, is the probability that a field is classified as infected by the model when FD was
 333 detected in that field. Similarly, the precision, also known as positive predictive value, is the
 334 probability that FD is detected in a field when it is classified as infected the model. A PR curve
 335 is then a plot of the precision (y-axis) and the recall (x-axis) for all possible thresholds between 0
 336 and 1. Predictive performance was summarised by the area under the PR curve (AUC). We also
 337 evaluated predictive performance at the district scale. Using the same 80% – 20% rule applied
 338 at the district scale, we evaluated the ability of the model to predict if the proportion (thereafter
 339 termed prevalence) of FD detection in a district exceeded a given threshold. Thresholds were

varied from 3% to 15%, and, for each threshold used, we calculated precision and recall for each partitioning.

Results

Model selection, goodness-of-fit and spatial field

The baseline model M_0 was fitted to 16 landscape scales ranging from 0 (only local variables considered) to 6000 metres. As a first step, we compared the 16 spatial models M_0 to their 16 non-spatial counterparts for which the spatial random effect $u(s_i)$ was removed. The AIC and DIC values of the spatial model were much lower at all scales (by at least 400 points), highlighting the importance of the spatial structure underlying FD detection in our data. The spatial model fitted with landscape variables at the 150 m scale was the best of the models fitted for both metrics considered (Fig. 2 A). Similar results were obtained with model M_1 . The second best landscape scale was 200 m for both models, parameter estimates being consistent between M_0 and M_1 at these two scales (Fig. S5). Thereafter, we then focused on the results obtained with a landscape scale of 150 metres, corresponding to a landscape area of 7.06 ha containing 6 fields on average (90% confidence interval [2 – 12] fields). Note also that, at this scale, the continuous explanatory variables were weakly correlated (Fig. S4).

The goodness-of-fit of the spatial model for a 150 m landscape scale was highly satisfactory. In particular, the coefficient of determination for the linear regression between the observed and adjusted prevalences of FD detection at the district scale was 0.99. Furthermore, this regression line is very close to the $y = x$ line, suggesting that the model explains most of the variability of the data at district scale (Fig. 2B).

Compared to the non-spatial model, the spatial model strongly reduces spatial autocorrelation as evidenced by comparing the spatial variograms of the residuals of both models (Fig. 2C). However, a weak spatial autocorrelation remains for distances of up to 3.6 km, the estimated range (95% confidence interval [3.1 – 4.2]). This is distance beyond which the correlation between the FD detection status of two fields becomes negligible. The shape of the Matérn function is itself instructive (Fig. 2D). In particular, the coefficient of correlation between the detection statuses of fields located up to 1 km apart is > 0.5 .

The distribution of spatial random effects provided evidence of a strong spatial pattern (Fig. 3A). Positive effects are clustered in the southern and western regions of the GDON des Bordeaux area, whereas negative effects are clustered in the north-western and the eastern regions. On the logit scale, the spatial random effect ranges from -1.33 (5% quantile) to 1.44 (95% quantile). These values correspond to odds ratios of 0.25 and 4.23, respectively. Moreover, the variation on the standard deviation is due to the clustering of FD detection cases over the study area (Fig. 3B).

Intra- and inter-annual effects on Flavescence doree detection

Inferences indicate that the probability of inspectors detecting FD during August is lower (odds ratio 0.24) than in September-October (Fig. 4A, last panel). The random walk used in model M_1 allows to refine the estimation of the effect of the weeks of field inspections and highlights a sharp increase from the first week of September (Fig. 5A). Moreover, the probability of FD detection was remarkably consistent for each of the five years surveyed except 2014, when it was somewhat significantly lower (Fig. 4A, last panel).

Flavescence doree risk factors at the field scale

Local variables describe the physical and agronomic characteristics of each field in the vineyard (Table 1). Their effects (posterior mean, 95% credible intervals and probability of being positive) are displayed in Fig. 4A. The corresponding odds ratio are obtained as the exponent of the posterior means. The probability of FD detection increases with the age of the plantation and its area but slightly decreases with planting density. No significant effect of the farming practice

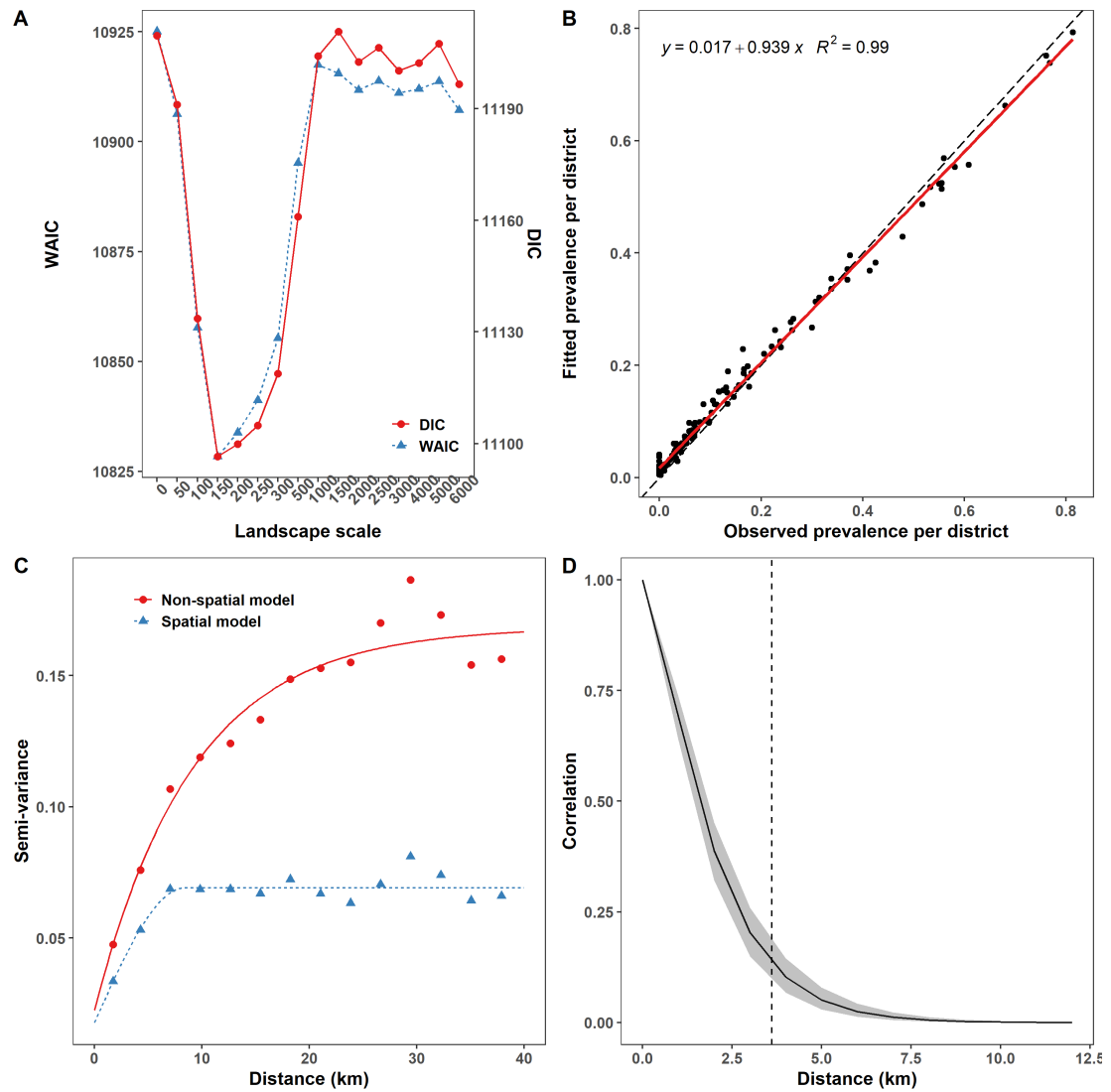


Figure 2: Comparison and fit of the spatial logistic models M_0 . A: Watanabe information criterion (WAIC) and Deviance information criterion (DIC) for the spatial model M_0 in 16 landscapes of increasing scales ranging from 0 m (only local explanatory variables considered) to 6000 m. B: Posterior mean estimate of the adjusted prevalence of FD detection against the observed prevalence aggregated at district scale. The fitted line between the adjusted and observed prevalences is shown in red and the $y = x$ line is shown in black. C: Sample variogram of the Pearson's residuals obtained for the spatial and non-spatial model M_0 at 150 m. D: The Matérn correlation function and its 95% credible band obtained with model M_0 at 150 m. The dashed vertical line indicates the estimated range.

was observed. The probability of FD detection decreases significantly with altitude (Fig. 4B). A 20 m increase in altitude is associated to an odds ratio of 0.72.

We analysed the effect of cultivar choice at field scale using Merlot, the most widespread cultivar in Bordeaux, as the reference (Fig. 4A, second panel from bottom). The odds ratio of FD detection in fields planted with Cabernet Sauvignon, Cabernet Franc and Muscadelle were, on average, 2.76, 2.29 and 2.08, respectively, compared to fields planted with Merlot. These differences were all highly significantly positive. Conversely, the odds ratio of FD detection in a field planted with Semillon were 0.76. No significant differences were found between Merlot and the cultivars Sauvignon and others (a class containing all the other minority cultivars).

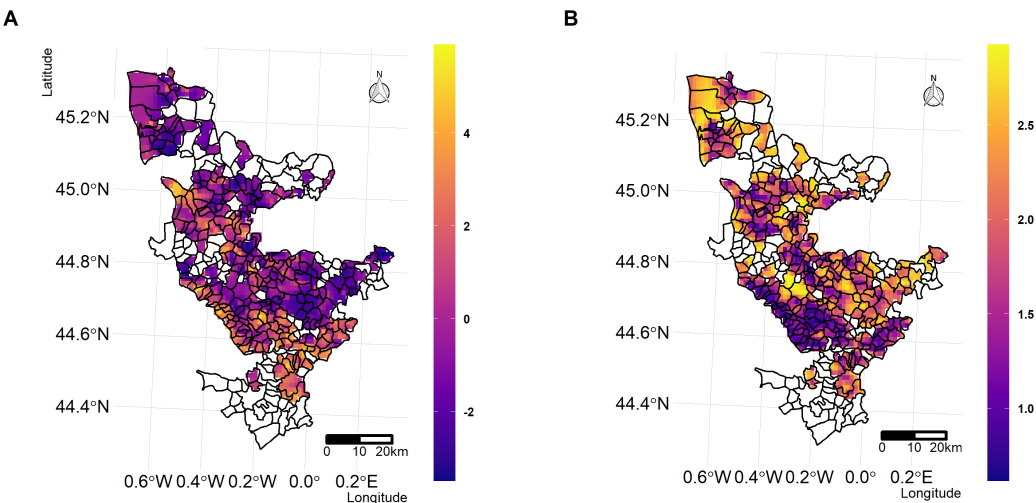


Figure 3: Spatial random effect estimated in the GDON des Bordeaux area for the baseline model M_0 for a landscape scale of 150 m. No data were available for the districts in white. A: Posterior mean of the spatial random effect. B: Posterior standard deviation of the spatial random effect.

397 Finally, the variable AOC was used as a proxy for the socioeconomic conditions of vine
398 cultivation, with AOC Bordeaux as the reference. For the six AOC levels considered, the
399 probability of FD detection was significantly higher only in the south-western part of the GDON
400 des Bordeaux area containing the AOCs Loupiac, Graves supérieures and Sainte-Croix-du-Mont
401 (Fig. 4A, fourth panel). The odds ratio of FD detection in these AOCs were 2.42 compared to
402 the AOC Bordeaux.

403 **Flavescence doree risk factors at the landscape scale**

404 The effects of the variables describing the composition and configuration of the landscape sur-
405 rounding a field within a landscape scale of 150 m are displayed in Fig. 4A (third panel). The
406 probability of FD detection increases with the proportions of forest and urban land and, more
407 slightly, with the proportion of organic fields. Conversely, the probability of FD detection de-
408 creases substantially with the proportion of vineyards. These effects are visualised in Fig. 4B.
409 For an increase of 20 percentage points in the proportion of forest, urban areas, or of fields with
410 organic practices, the odds ratio of detection are 1.28, 1.34 and 1.13, respectively whereas a 20
411 percentage points increase in the proportion of vineyards is associated to an odds ratio of 0.84.

412 We also tested the effect of the varietal composition of the landscape throughout the pro-
413 portion of Merlot area in the vineyard. The probability of FD detection decreases substantially
414 with increases in the proportion of fields planted with Merlot (Fig. 5 B). An increase of 20
415 percentage points in the proportion of fields planted with Merlot, for a landscape scale of 150
416 m is associated to an odds ratio of 0.93.

417 **Predictive performance of the model**

418 We evaluated the ability of the model to predict the FD detection status of individual fields. The
419 corresponding PR curve indicates a moderate predictive performance (Fig. S6A), as summarised
420 by the AUC of 0.59 (95% credible interval [0.54, 0.64]). The predictive performance was better
421 at district scale. Specifically, we evaluated the ability of the model to predict if the prevalence
422 of FD detection in a district exceeds a given threshold (Fig. S6B). Recall decreases with the
423 threshold, whereas precision remains constant. For a threshold of 10%, the posterior mean recall
424 (*i.e.* sensitivity) is 89%. This is the probability of the model correctly classifying a district as

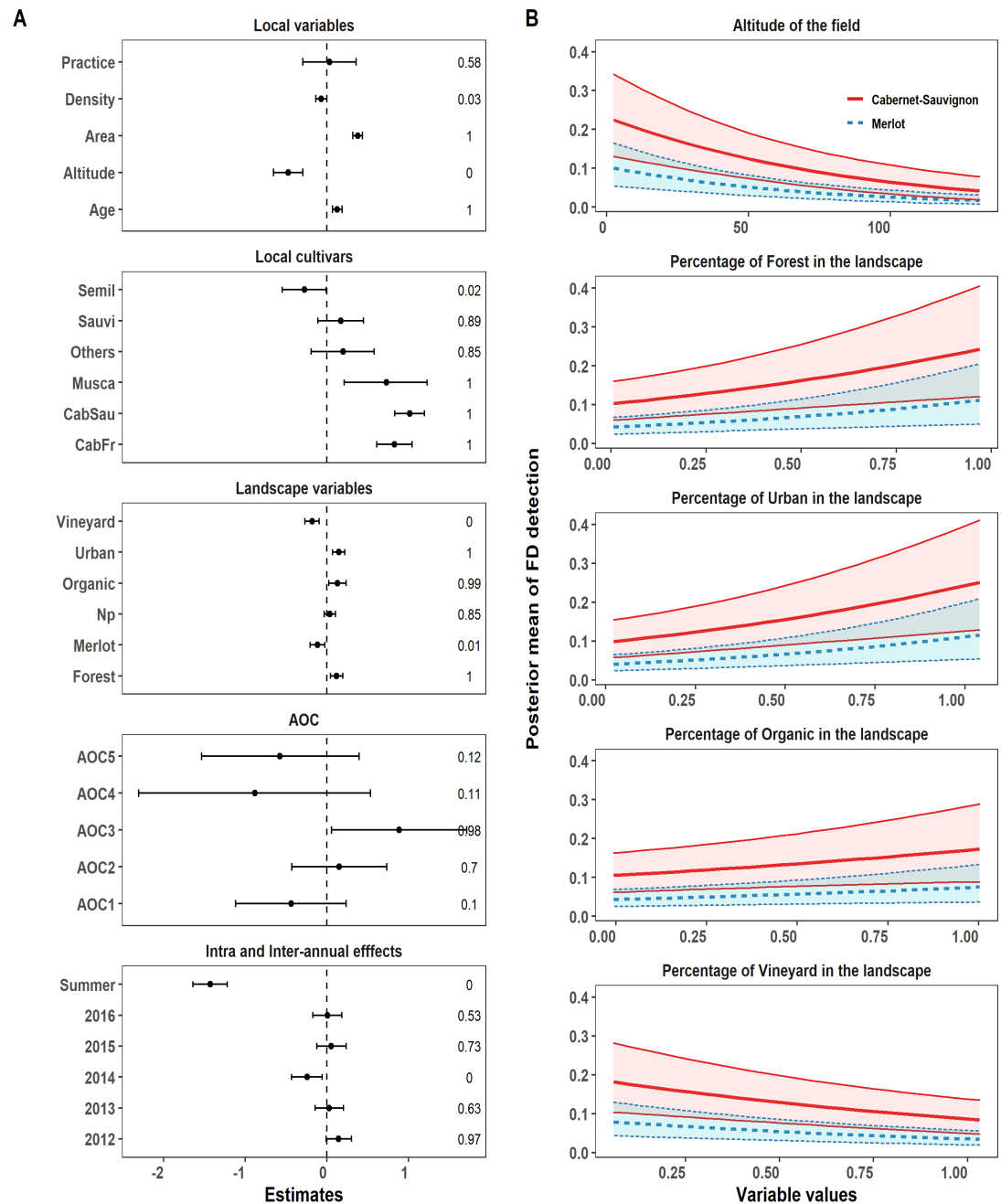


Figure 4: Estimation of the parameters of the baseline model M_0 for a landscape scale of 150 m. A: Fixed and random effects of the local and landscape explanatory variables. For each variables, the posterior mean (dots) and 95% credible intervals (solid lines) are displayed, together with the posterior probability of the effect being positive. The dashed line corresponds to the value 0. B: Effects of altitude, proportions of forest, urban area, organic fields and vineyard in the landscape on the probability of FD detection for the two most widespread cultivars (Merlot and Cabernet Sauvignon) in vineyard plots under conventional practices (mean probability and 95% credible bands). The effects were estimated by fixing all other numerical variables at their mean values, the variable AOC to Bordeaux, the variable Practice to conventional and considering a sampling period during the autumn. All variables and their levels are described in Table 1. We dropped the "L_" from the landscape variables to lighten notations.

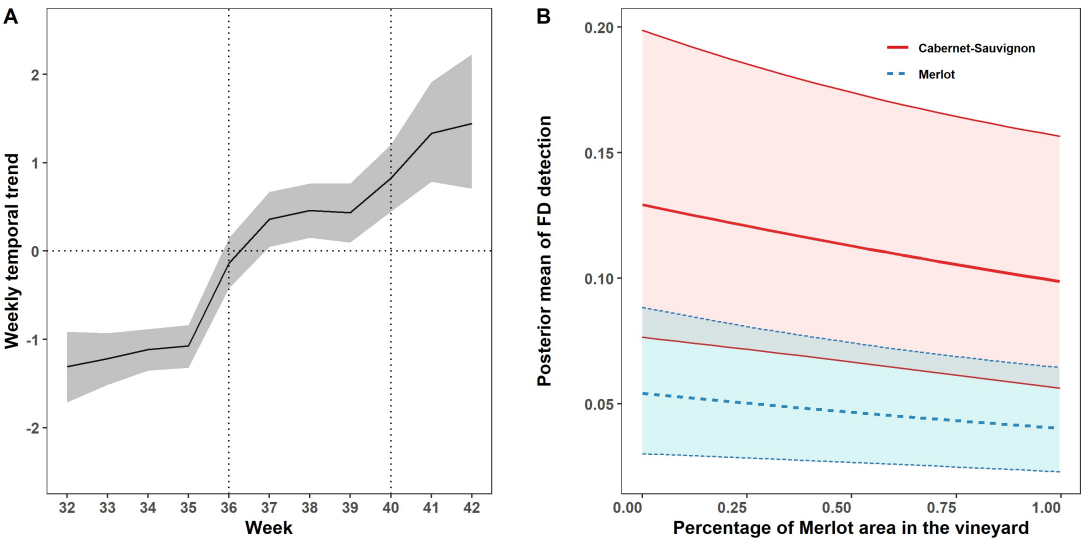


Figure 5: Effects of the week of field inspection and of the cultivar composition of the landscape on FD detection, for a landscape scale of 150 m. A: Effects of the week of field inspected on FD detection as estimated with model M_1 at 150m. In model M_1 , the fixed effect variable *Season* is replaced by a random walk of order 1 indexed by weeks. The mean standardized temporal effect (from week 32, second week of august, to week 42, third week of october) is displayed along with its 95% credible band. B: Effect of the proportion of Merlot in the landscape on the probability of FD detection, using the model M_0 for the two most widespread cultivars (Merlot and Cabernet Sauvignon) in vineyard plots under conventional practices (mean probability and 95% credible bands). The effects were estimated by fixing all other numerical variables at their mean values, the variable AOC to Bordeaux, the variable Practice to conventional and considering a sampling period during the autumn.

425 having a prevalence of FD detection exceeding 10% of the fields. The posterior mean precision
426 (*i.e.* positive predictive value) is then 50%. This is the probability that prevalence of FD
427 detection exceeds 10% in a district when the district was classified as such by the model.

428 **Discussion**

429 In this study, we used species distribution models fitted in a Bayesian framework to identify
430 key biotic and abiotic factors driving the detection and epidemiology of Flavescence dorée, a
431 major quarantine disease damaging European vineyards. To this end and based on our eco-
432 logical knowledge of the disease, we integrated data from different sources (*e.g.*, FD survey
433 data, agricultural management and environmental explanatory variables) to build a quantita-
434 tive understanding of FD infection. Our study makes use of a large spatial dataset gathering
435 34581 observations of the infection status of vineyard plots. It relies on an accurate moni-
436 toring of symptoms realized by trained inspectors completed with molecular detection of FD
437 phytoplasma on symptomatic plants performed by accredited laboratories. Due to the costs
438 of laboratory testing, no tests are conducted on leaves from non-symptomatic plants, opening
439 possibly the way to false negative results due to cryptic infection ((Parnell et al. 2017)). To this
440 respect, our analysis highlights the importance of sampling period on FD detection (variable
441 *Season* in model M_0 and *Week* in model M_1). Both models indicate that the probability of
442 FD detection substantially increases after week 36 (first week of September, Figs. 4A, 5A). The
443 visual identification of FD symptoms is then much easier as this period is the ideal time for the
444 expression of certain specific symptoms, such as grape shrivelling and the non-lignification of
445 canes (Tramontini et al. 2020). Accordingly, from September onwards, the risk of false negative
446 results is low. Furthermore, recent studies show that non-symptomatic plants sampled at the end

of the summer in fields with high FD prevalence were actually not infected by the phytoplasma (Eveillard et al. 2016, Ripamonti et al. 2020). In particular, the study of (Eveillard et al. 2016) was conducted in the Bordeaux region over hundreds of Cabernet-Sauvignon and Merlot plants, the two most widespread cultivars in our dataset (76 % of the samples). Moreover, inspectors collected in each field a wide diversity of symptomatic leaves possibly related to phytoplasmas. Among the 4554 molecular detection performed in as many fields, 47% were positive to FD. Accordingly, based on these premises, we are pretty confident that our analysis provide important information on how local farming practices and landscape context affect the probability of FD infection. At local scale, our results reveal in particular that the probability of detecting infected fields increases with field area, the age of the plantation, and with specific cultivars (i.e., Cabernet Sauvignon, Cabernet Franc and Muscadelle). At the landscape scale, the probability of FD infection increases with the proportion of susceptible cultivars as well as with the proportion of forest, urban land and organic farming but decreases with the proportion of vineyard.

At field scale, cultivar choice, but not other viticultural practices considered, affected the probability of field infection. Cultivar choice was a major determinant of FD infection. No major resistance gene effective against FD is currently available, but substantial differences in cultivar susceptibility have been reported (Oliveira et al. 2019, Eveillard et al. 2016). In the GDon des Bordeaux region, the most frequently grown cultivars are Merlot (in 61% of the fields analysed), Cabernet Sauvignon (in 15% of the fields), Cabernet Franc and Semillon (in 7% of the fields each). Previous studies involving controlled inoculations in greenhouses or monitoring in experimental plots have shown that Cabernet Sauvignon and Cabernet Franc are more susceptible to FD than Merlot, displaying a higher incidence of the disease, more severe symptoms and higher rates of phytoplasma multiplication (Eveillard et al. 2016). Our results therefore confirmed this ranking in production situations for the first time. They also suggest that Semillon has low susceptibility to FD, whereas Muscadelle is highly susceptible to FD. In addition, our analyses revealed that the altitude substantially affects FD infection : the higher the altitude of the field, the lower its probability of infection (Fig. 4B). This effect fits the empirical observation that, in the region studied, FD is more likely to occur in wetland areas, which are frequently found at lower altitudes, typically close to rivers. The slightly lower goodness-of-fit obtained by replacing altitude by an wetland index (Merot et al. 2003) have led us to favour this variable which is, also, easier to access. Identifying the underlying biological factors deserve further studies.

Our study reveals that field infection by FD were also affected by the landscape context within the 150 – 200 m surrounding focal fields. The best fit were obtained with the landscape scale of 150 m, followed by 200 m. These landscape extent corresponds to areas ranging from 7 ha (150 m scale) to 12.5 ha (200 m scale) and contains, in average, 6-10 fields in the study area (the mean field area is 0.76 ha). This short range may result from the low dispersal capacity of the vector of the FD phytoplasma. Indeed, 80% of *S. titanus* adults disperse within 30 m of their source, although long-range dispersal over distances of up to 330 m has occasionally been observed (Lession et al. 2014). Several effects of landscape composition were identified (Fig. 4B). Increases in the proportions of two generic land-use classes, forest and urban land, increased the probability of FD infection. The interpretation of such effects is not straightforward because generic land-use classes provide only an imperfect description of the presence of host habitats involved in epidemiological dynamics (Vanwambeke et al. 2019). In our case study, it is tempting to consider the positive effect of the proportion of forest as favoring wild reservoirs of FD, such as wild alders and *Clematis* sp. However, recent studies have demonstrated that the transfer of the phytoplasma from these plants to grapevine is rare, so their impact as chronic reservoirs is likely to be minimal (Filippin et al. 2009, Malembic-Maher et al. 2020). The hypothesis involving vines, whether abandoned or cultivated, is more likely. Indeed, abandoned *Vitis* sp. plants in forest margins and outskirts constitute an important source of FD inoculum but also of insect vectors (Lession et al. 2014, Ripamonti et al. 2020). To this respect, it could be interesting to refine landscape description in order to investigate the role of the size of forest patches as smaller patches offer more surface on their outskirts compared to larger ones. Similarly, inventories and cartography of *Vitis* sp. in two districts of Bordeaux area have shown that cultivated vines in individual gardens or uncultivated *Vitis* sp. in wasteland are frequent

in urbanized areas neighboring vineyards (data not published). They could also constitute an important source of FD inoculum and insect vectors, especially as residents and gardeners have little awareness of the problem of FD. Alternatively, it is also likely that fewer insecticide treatments are performed by farmers in vineyards close to private houses or public facilities but also by amateur gardeners, which might also explain this result.

In addition, we identify that the probability of FD infection decreases with the proportion of vineyards in the landscape but also with planting density. This effect, which appears counter-intuitive at first glance, may come from several processes. It may reflect a direct dilution effect of the available inoculum or of the vector population within larger cultivated vineyards. In line with this hypothesis, (Delaune et al. 2021) have recently shown that univoltine pest species such as *S. titanus* are negatively correlated with the host crop area in the landscape during the ongoing growing season. This effect may also reflect that larger cultivated vineyards can benefit from a more homogeneous insecticide protection (Meehan et al. 2011). Alternatively, it might be an indirect effect mediated by the economic value of vineyards. Indeed, areas with highly valued vineyards are generally more specialised, with more financial incentives for farmers to control the infection (Ay and Gozlan 2020).

Interestingly, effects of cultivars detected at the field scale were also found at the landscape scale suggesting up-scaling of these effects that may be mediated by dispersal abilities of *S. titanus* populations. We show that the probability of FD infection decreases with increasing proportions of Merlot (Fig. 5B). Several studies have shown that the landscape composition in plant species (*i.e.* between-species diversity) affect plant disease epidemiology (Plantegenest et al. 2007, Meentemeyer et al. 2012). However, studies on the effect of varietal (*i.e.* within-species diversity) composition at the landscape scale are much rarer and mainly based on theoretical analyses (Rimbaud et al. 2021). Few empirical studies have suggested such an effect and they mainly considered regional or national scales (Priestley and Bayles 1980, Finckh and Wolfe. 2017, Papaix et al. 2011)). By contrast, our study suggests an effect of the varietal composition of the landscape at an intermediate spatial scale, typically over an area of 7 – 12 ha, containing about 6-10 fields. Finally, while no effect of farming practices (organic *versus* conventional) was detected at the field scale, our result suggest that the probability of FD infection may slightly increase with the proportion of organic fields in the landscape (Fig. 4B). Note first that this variable indicates if organic practices are used at farm scale rather than for each individuals fields (fields being attached to their farm in our study). It could possibly explain this mismatch between the results at local and landscape scales. The positive effect at landscape scale could suggest a lower efficacy of vector control in organic systems. The natural products used in organic vineyards to control *S. titanus* are as effective as synthetic insecticides in controlling nymphs but less effective in controlling adults, the most dispersive instar (Tacoli et al. 2017, Guesberti et al. 2008, Chuche and Thiéry 2014). Combined with the stricter conditions governing field applications, it might account for this effect. However, this hypothesis needs further investigations as we did not have the information about the number of mandatory treatments actually realized. Moreover, other practices used in conventional and organic viticulture (*e.g.* soil tillage, cover crops or canopy management) that were not considered here might have affected the spread of FD and the dynamics of its vector (Muneret et al. 2018).

The local and landscape factors identified can be used to improve vineyard surveys, thereby decreasing the economic and environmental impacts of managing this disease. The use of the model as a predictive tool is potentially interesting for the targeting of new districts in which future surveys are most likely to be valuable, as suggested by its to the ability of the model to discriminate districts with prevalence of FD detection exceeding 10% of the fields (Fig. S6B). Out-of-sample predicted probabilities of FD presence could be used in a benefit-cost framework in order to determine the optimal size of future surveys or to improve the current control policy (Ay and Gozlan 2020). The grubbing costs associated to FD presence would be derived from farm's gross margins or vineyard prices and compared to the monitoring costs from professional organisations such as GDON. Moreover, the Bayesian method used (INLA) allows uncertainty to be quantified and visualised in risk map outputs. This is important for fostering dialogue with stakeholders and policy-makers. However, and importantly, using the model as a predictive tool must firstly be restricted to make interpolation to unsampled sites within the GDON

des Bordeaux area and not as a tool able to make prediction in remote vineyards. Moreover, given the importance of the spatial effects (as discussed below), the model has also to be fitted again annually by taking into account new FD survey in order to update the spatial field to the current state of the epidemic in the area. A short-term operational output of the result to improve FD monitoring concerns the effect of the sampling period. It suggests that the harvest period in Bordeaux (September) is a more favourable period for maximising the efficacy of vineyard surveys (Fig. 4A). Note that part of the size of this effect could also be explained by the GDON internal organization of field surveys as the last weeks of survey, mostly after week 38 (*i.e.* over 2.8% of the sample), are realized by more experienced teams of inspectors while sometimes targeting districts closer to previously identified infected fields. This result strongly suggests that equipping grapevine harvesting machines with cameras and automatic symptom recognition tools could greatly improve FD detection, especially since harvesting machines travel through a large part of the vineyard every year. Recently, the use of convolutional neural networks has shown promise for detecting FD symptoms from in-field images (Boulent et al. 2020). Finally, our results may have important consequences for the management of vineyard landscapes echoing the land sparing / land sharing debate (Phalan et al. 2011). Our study suggests that promoting landscapes with large amount of vineyards and lower proportions of forests limit FD infections while maintaining semi-natural habitats in such landscapes are known to be of major importance for biodiversity conservation or mitigation of climate change effects (Rusch et al. 2022, Barbaro et al. 2021). Policy makers and land use planners may therefore include such issue if they are to design multifunctional vineyard landscapes.

In addition to the effects of the fixed factors, a large proportion of the biological process of interest was captured by the spatial effects included in our models. In our case study of more than 35000 observations, the INLA method proved effective for implementing complex Bayesian hierarchical models, taking spatial autocorrelation into account. The spatial component, as revealed by the Matérn function (Fig. 2D), had a strong effect, suggesting that some biotic (*e.g.* specific cultural practices) and abiotic (*e.g.* climatic variables) factors are missing or misspecified (*e.g.* linear terms where nonlinear terms are required) (Elith and Leathwick. 2009). However, the strong spatial pattern observed may also be the result of several epidemics developing simultaneously at local level, over small scales, consistent with the short dispersal distances of the vector, whereas the distribution model is based primarily on a static vision of the disease risk factor in an environment at equilibrium (Elith and Leathwick. 2009). This assumption does not hold for invading pathogens (Purse and Golding 2015). In recent years, the idea of coupling niche modelling with spatially explicit models of disease dispersal to provide better information about potential disease spread and ultimately improve epidemiological surveillance strategies has been explored by several authors (Purse and Golding 2015, Parnell et al. 2017, Martinetti and Soubeyrand 2019, Rimbaud et al. 2018, Cuniffe et al. 2016, Hyatt-Twynam et al. 2017, Adrakey et al. 2017). This approach merits implementation in this context, to improve prediction accuracy at field scale. This implementation could make use of existing models of FD epidemics at field scale (Lesson et al. 2015). The insights gained from our correlative study could be incorporated into mechanistic models of epidemic spread (Meentemeyer et al. 2011, Hartemink et al. 2011). Moreover, it should also be recognised that correlative models, which can easily take into account large numbers of predictors, sometimes outperform mechanistic models and can have a major impact on policy decisions (Leach and Scoones. 2013). From this point of view, it may also be useful to explore the rapidly developing research fields of machine learning and data mining. These methods have been successfully applied to the surveillance of *X. fastidiosa* (Martinetti and Soubeyrand 2019). This possibility is particularly attractive, given that machine learning methods have recently been adapted to take spatial autocorrelation into account (Georganos et al. 2021).

Authors' contributions

HA, SM and FF conceived the ideas and designed methodology; HA, SM, LR and FF collected the data; HA, LR, AR, JSA and FF analysed the data; HA and FF led the writing

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877 Supplementary Materials

878 **Supplementary Materials for "Field and landscape risk factors impacting Flaves-**
 879 **cence dorée infection : Insights from spatial Bayesian modelling in the Bordeaux**
 880 **vineyards"** by Adrakey HK., Malembic-Maher S., Rusch A., Ay JS., Riley L., Ramalanjaona
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882 Text S1: Attributing CVI variables to field sampled by GDON

883 The Flavescence Doree (FD) dataset analysed provides the FD detection status of 34581 fields
 884 sampled between 2012 and 2016. During their field trips, trained inspectors delimit in a GIS
 885 database with a spatial polygon which they identify as an homogeneous area of vine cultivation,
 886 thereafter termed GDON field. GDON fields are supposed to share the same year of plantation,
 887 density of plantation and cultivar. Unfortunately, GDON inspectors do not have access to these
 888 variables. However, they are available in the CVI (Casier Viticole Informatisé), a database
 889 created by the French directorate general of customs. The CVI gives a comprehensive history
 890 of each legal piece of land referenced in an administrative land register (Cadastre in French).
 891 However, matching these information to GDON fields is not always straightforward as the
 892 polygons drawn by GDON' inspectors do not necessarily strictly overlap those of the CVI.
 893 Initial work (Table S1) focused on mapping information from the CVI to each GDON field -
 894 enabling extraction of information such as the field's age and the majority cultivar grown as
 895 shown in Fig. 1.

896 Most of the CVI fields are associated to a single agronomic field as defined by an unique set
 897 of values for the year and plantation, plantation density and cultivar. Out of the 34581 GDON
 898 fields, 15204 are associated to a single agronomic field. However, a CVI field can also be divided
 899 into several agronomic fields. The difficulty here is that agronomic fields are not associated to
 900 their own spatial polygon. Only their areas are known. A case by case study is then necessary
 901 in order to assign the more likely values of year of plantation, plantation density and cultivar to
 902 each GDON field. The cases are detailed in Table S1 along with their frequency of occurrence
 903 among the 34581 GDON fields analysed. For year of plantation and plantation density, the
 904 process consists in assigning to each GDON field metrics corresponding to mean weighted by
 905 the surface of agronomic fields of the CVI field where the GDON field of interest is located. For
 906 example an estimate of the planting year in the case 3 (Table S1) is derived as weighted mean of
 907 all agronomic fields planting years weighted by their respective areas. Similarly, for the cultivar,
 908 the process consists in assigning to GDON field the cultivar grown on the largest agronomic
 909 field of the CVI. This process occurs in 11927 GDON fields (Table S1, cases 4 to 6). In these
 910 cases, the largest agronomic field occupy on average 63% of the CVI field (quantile 10% = 0.43,
 911 quantile 90% = 0.85).

912 Also, the spatial polygons of the GDON fields do not always exactly match those of the
 913 CVI fields as they are drawn independently (initially taken from an ortho-photograph and later
 914 adjusted during inspections). Accordingly, noting that GDON fields are in general smaller than
 915 CVI fields, we assigned the information issued from a CVI field to a GDON field if its centroid
 916 lies within the polygon of the CVI field.

Table S1: Columns 1-6: Summary of scenarios resulting from the information provided by the CVI and their interpretation. Column 7: The number of GDON fields to which CVI legal fields belonging to each scenario were assign to, with the sum corresponding to the size of the population under study.

Cases	Sum of agromomic field area within the administrative field	Number of agromomic fields within the administrative field	Year of planting	Cultivars	Interpretation	Number of occurrence in the Gdon data (2012-2016)
1	$\leq$ cadastral land area	1			Administrative field having only one agromomic field and with same or less area compared to the crop area	15203
2		≥ 1	Identical	Identical	Administrative field having multiple agromomic fields with the same planting year and cultivars and their total area is smaller than the cadastral one	1373
3			Different	Identical	Administrative field having multiple agromomic fields with the same cultivars but different planting year and their total area is smaller than the cadastral one.	6105
4			Identical	Different	Administrative field having multiple agromomic fields with different cultivars but the same planting year and their total area is smaller than the cadastral one	895
5			Different	Different	Administrative field having multiple agromomic fields with different cultivars, different planting year and their total area is smaller than the cadastral one	10976
6	$\geq$ cadastral land area	≥ 1	Different	Different	Administrative field having multiple agromomic fields with the same planting year and their total area is greater than the cadastral one	19
7			Identical	Identical	Administrative field having multiple agromomic fields with the same planting year and cultivars and their total area is greater than the cadastral one	8
8		1	Identical	Identical	Administrative field having only one agromomic field and with same or bigger area compared to the crop area	2

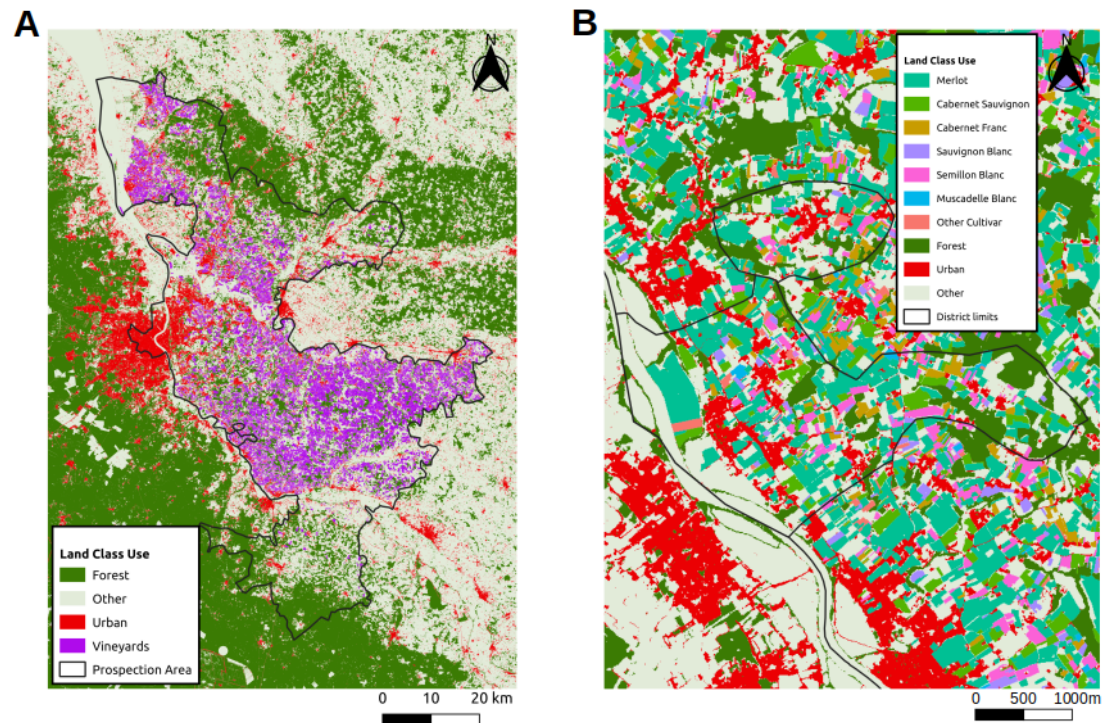


Figure S1: Land cover map for the year 2016. The land cover maps used to characterize the landscape surrounding each field are raster maps with a 10-meters resolution. Each pixel was classified into 1 of 11 categories. The first 8 categories describe the vineyard field : cultivars (categories 1 to 7) and cultural practice (category 8 : organic or conventional ; not illustrated here as it overlap with the cultivars). The 3 last categories corresponds to other land use : forest, urban or other land use. The class forest merges the classes 31 (Broad-leaved Forests) and 32 (Coniferous Forests) of the OSO map for 2017. Similarly, the class urban merges the classes 41 (Continuous Urbain Fabric), 42 (Discontinuous Urbain Fabric) and 43 (Industrial and Commercial Unites) of the OSO map for 2017. The panel A illustrates the 4 classes of land use vineyard, forest, urban and others over the whole area monitored by GDON des Bordeaux. The panel A illustrates the 4 class of land use vineyard, forest, urban and others over the whole area survey by GDON des Bordeaux. The panel B also details in a smaller area the cultivars used in the vineyard plots.

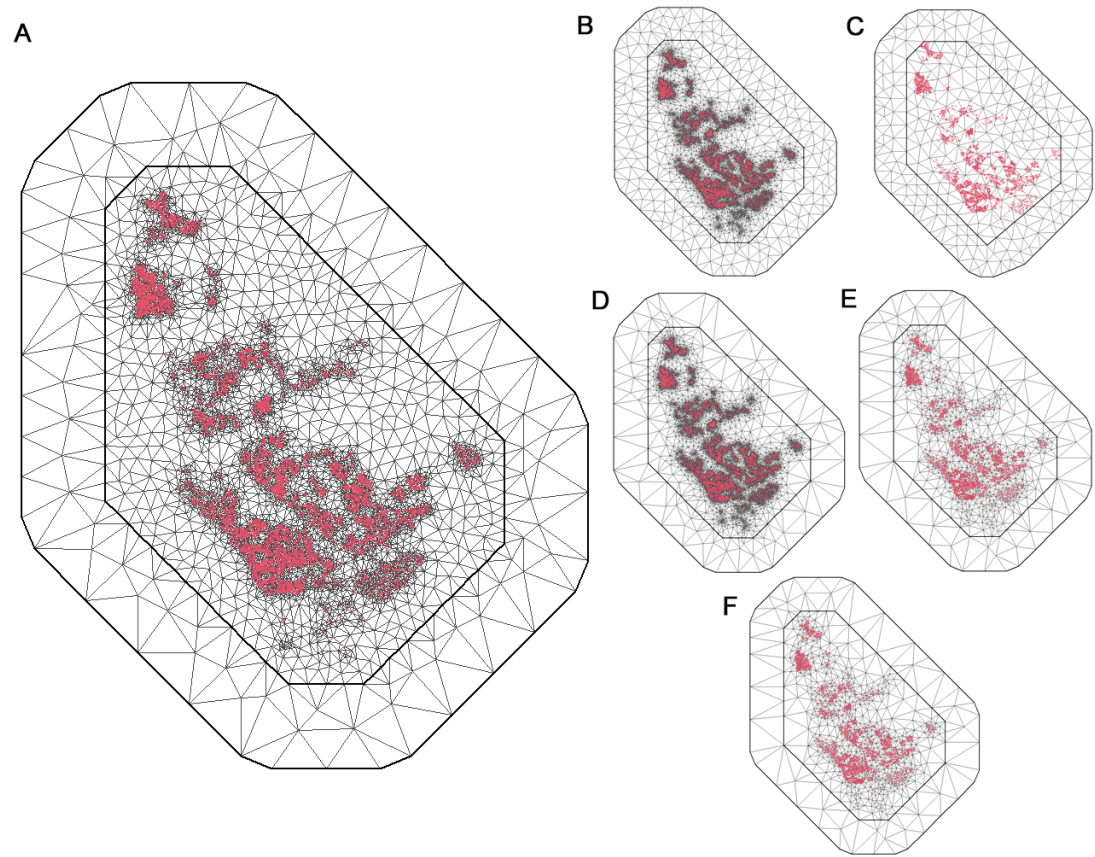


Figure S2: Boundary of the study region and the mesh created to approximate the solution of the SPDE to obtain the spatial model. A thick line separates the outer offset from the inner offset and the boundary of the study region. The mesh is extended beyond the study area to avoid a boundary effect. The meshes differ in the parameters max.edge (the largest allowed triangle edge length) and cutoff (the minimum allowed distance between points). A more precise mesh is expected to provide a better estimation of the spatial effect (the prediction will be smoother) but this comes at the cost of longer computational times. A: $\text{max.edge}=\text{c}(5,25)$ and $\text{cutoff}=0.25$. This is the mesh used in the main text. B: $\text{max.edge}=\text{c}(10,10)$ and $\text{cutoff}=0$. C: $\text{max.edge}=\text{c}(10,10)$ and $\text{cutoff}=10$. D: $\text{max.edge}=\text{c}(50,50)$ and $\text{cutoff}=1\text{e-}12$. E: $\text{max.edge}=\text{c}(75,75)$ and $\text{cutoff}=1$. F: $\text{max.edge}=\text{c}(25,50)$ and $\text{cutoff}=1$. Values are in kilometers. The corresponding number of vertices associated from panel A to F are 8845, 58209, 317, 58078, 1296 and 1296, respectively.

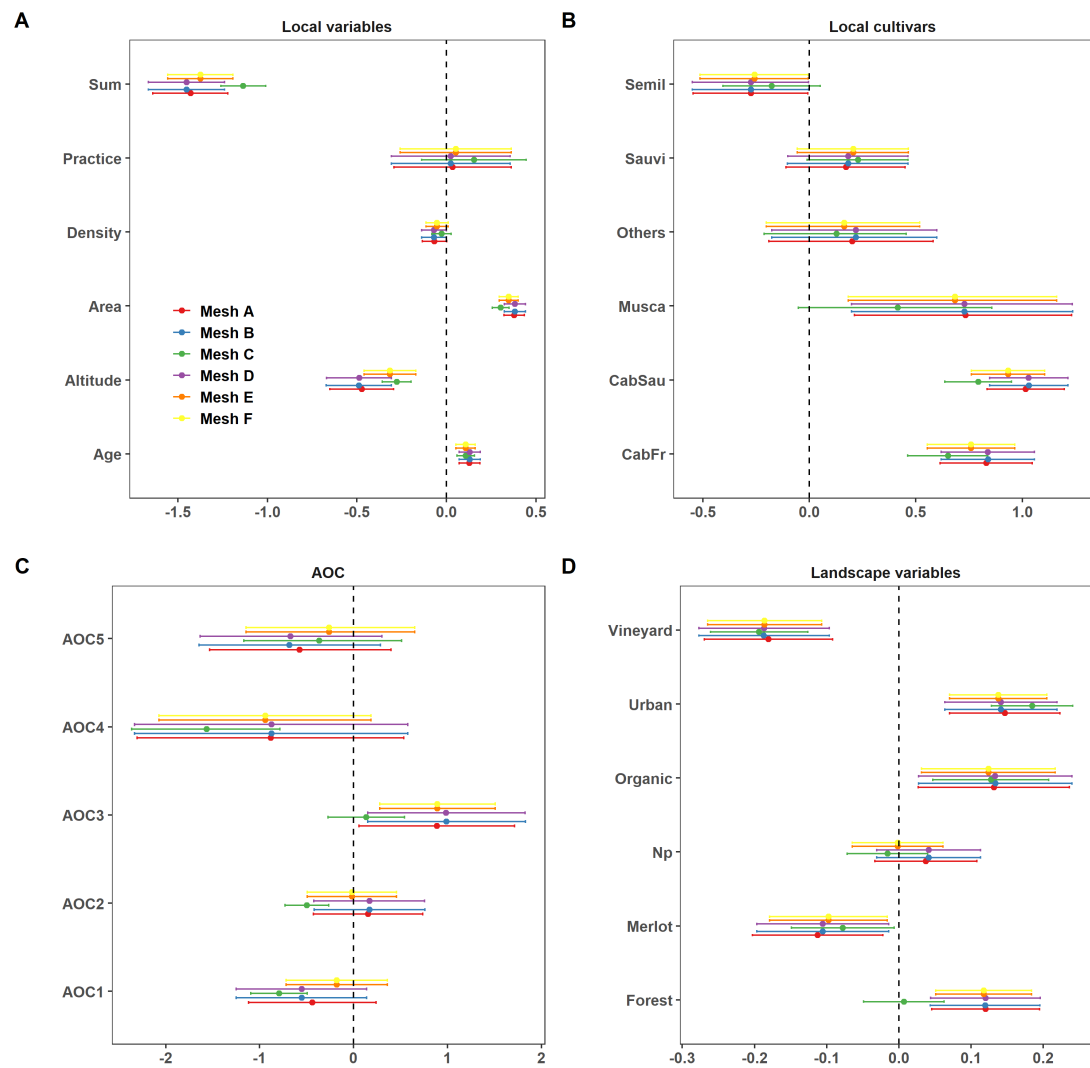


Figure S3: Estimation of the parameters associated to fixed effects of the baseline model M_0 (at a landscape scale of 150 m) for six meshes. The panels displayed the posterior mean (dots) and 95% credible intervals (solid lines) of every local covariates, cultivars, AOC and landscape covariates for the six meshes used to define the spde model (Fig. S2). Mesh A corresponds to the mesh used to derive results in the main text (Fig. S2A). Dashed line corresponds to the values 0. For clarity, we dropped the "L_" from the landscape variables.

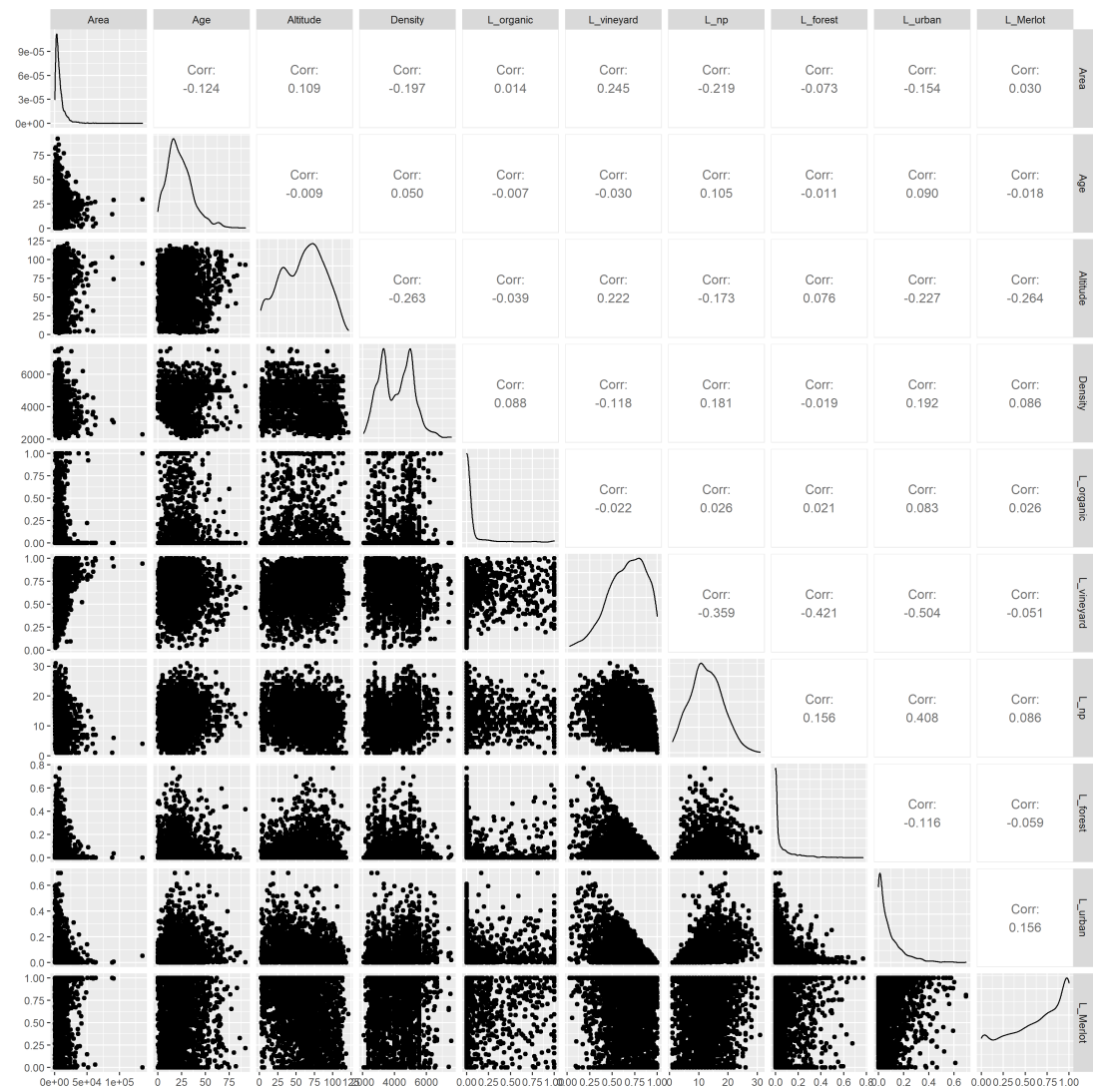


Figure S4: Correlation matrix between the continuous explanatory variables used in the model M_0 at the landscape scale of 150 m. A random sample of 2500 out of the 34581 observations is used to draw the pairwise scatter plots.

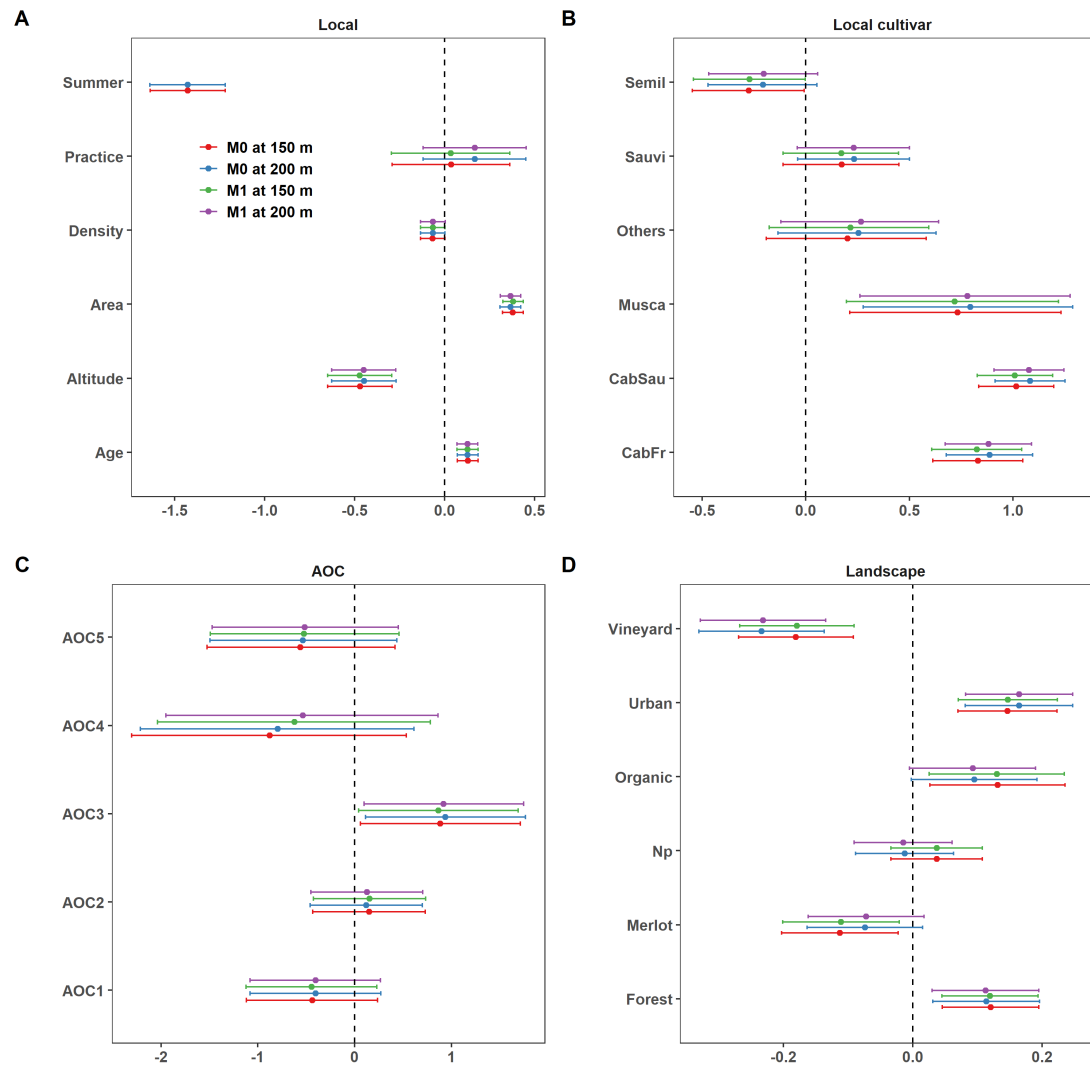


Figure S5: Estimation of the parameters associated to fixed effects of models M_0 and M_1 at the landscape scales of 150 m and 200 m. The landscape scale of 150 m is associated the lowest AIC and DIC for both models M_0 and M_1 , followed by the landscape scale of 200 m. The panels displayed the posterior mean (dots) and 95% credible intervals (solid lines) of every local covariates, cultivars, AOC and landscape covariates for the four combinations (two models times two scales). Dashed line corresponds to the values 0. For clarity, we dropped the "L_" from the landscape variables

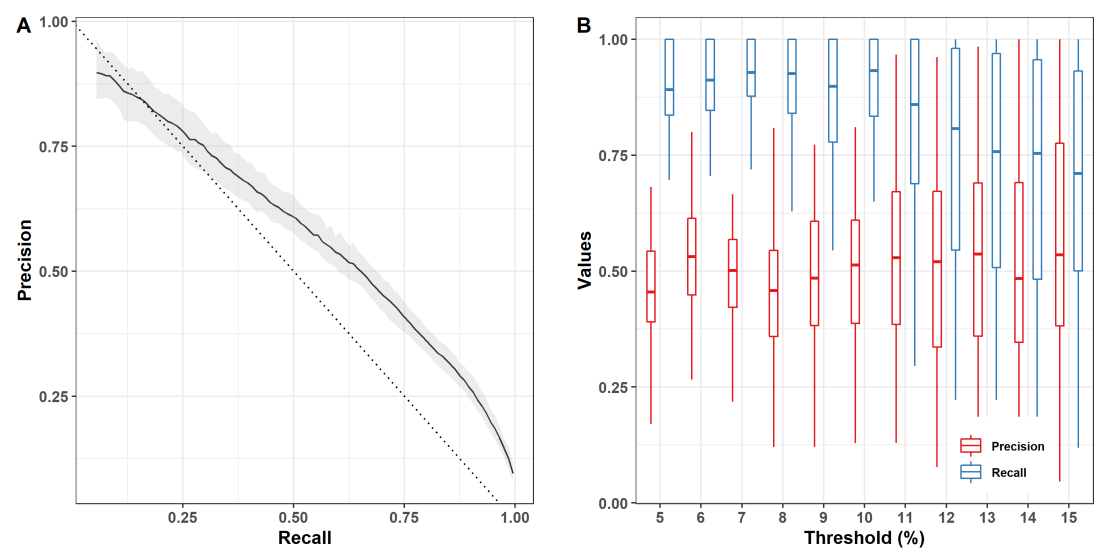


Figure S6: Predictive performance of the baseline model M_0 in a landscape scale of 150 m. A: Predictive performance at field scale, as estimated by the mean precision-recall curve (in black) and its 80% credible band (in grey), based on 50 cross-validation partitions. B: Predictive performance at district scale, as estimated by the ability of the model to predict if the prevalence of FD detection in a district exceeds a threshold. For each threshold, the marginal posterior distribution of precision and recall were obtained from 50 cross-validation partitions.